MILITARY SPECIFICATION

COUNTER, ELECTRONIC, DIGITAL READOUT

26.5 GIGAHERTZ (GHZ) CONTINUOUS WAVE (CW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state automatic microwave frequency counters hereinafter called equipment.

1.2 Classification. Equipment covered by this specification will be as defined in MIL-T-28800C and herein for type III, class 5, style E, and color R.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596/17C	Connector, Plug, Electrical, General Purpose, Grounding, General Grade 2 Pole, 3 Wire, 15 Amperes, 250 Volts, 50/60 Hertz

MILITARY

MIL-N-7513	Nomenclature Assignment, Contractors Method for Obtaining.
MIL-Q-9858	Quality Program Requirements.
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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MIL-S-19500	Semiconductor Device, General Specification for.
MIL-C-28777	Cable Assembly, electronic Test Equipment (3 Wires, 125 and 250 Volts AC and 28 Volts DC) Grounding Plug Connector General Specification for.
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for.
MIL-P-55110	Printed Wiring Boards.
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, for Aerospace Applications.

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

MIL-STD-210	Climatic Extremes for Military Equipment.
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1519	Test Requirements Document, Preparation of.
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapon Systems.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE-488-1978	Standard Digital Interface for Programmable Instrumentation.

(Application for copies should be addressed to the IEEE Standard Sales, 345 East 47th Street, New York, NY 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Sample submissions. When required in the contract or order or solicitation, sample unit(s) shall be furnished for initial production article inspection and approval.

3.2 Safety. The equipment shall comply with the type III safety requirements specified in MIL-T-28800C (see 4.5).

3.3 Parts, materials and processes. Unless otherwise specified herein, the equipment shall comply with the type III parts, materials, and processes specified in MIL-T-28800C.

3.3.1 Restricted materials. Equipment shall comply with the restricted material requirements of MIL-T-28800C including the requirement for mercury or radioactive materials and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing. No dissimilar metals should be used as defined in MIL-STD-889.

3.3.2 Input and output signal connectors. Except for the logic interface, the low impedance input port(s), and the extended frequency port(s), all input and output ports shall be series BNC female. The logic interface connector shall be in accordance with 3.12.10.

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements specified in MIL-T-28800C.

3.4.1 Solid-state construction. The equipment shall be of solid-state modular miniaturized construction. Electron tubes shall not be used. The semiconductors used within the counter must comply with MIL-S-19500.

3.4.2 Mainframe plug-in concept. Equipment utilizing externally accessible, externally removable subassemblies (mainframe plug-in concept) shall not have plug-in(s) which are removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to this specification.

3.4.3 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible. All potentiometer controls shall have at least a 4.7 radian (270 degrees) rotational range.

3.4.3.1 Front panel controls. Controls for the following functions shall be located on the front panel.

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a. Standby-operate (or off-on, provided that power to oven is applied when counter is connected to an active ac power source).

b. Sample rate.

c. Resolution.

d. Reset.

e. Self check.

f. Input port selection.

3.4.4 Accessibility. The equipment shall be constructed so that:

a. Subassemblies and chasis components can be removed without removing any hard wired printed circuit cards, subassembly, or component.

b. Adjustments can be made without removing any component, printed circuit card or subassembly except the use of extender cards is permitted.

c. Printed circuit cards can be removed without the need to unsolder cables and interconnecting wiring. Printed circuit cards (mother boards) designed primarily to distribute power and signals to other printed circuit cards (daughter boards) are excluded from this requirement. When such boards are used, they shall be accessible from both sides with minimum disassembly. Printed circuit cards shall be manufactured in accordance with MIL-P-55110.

3.4.5 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies shall not be used.

3.5 Electrical power sources and connections.

3.5.1 Power sources. The equipment shall comply with the type III, 50 to 60 hertz (Hz) single phase 115/230 volt (V) and 400Hz single phase 115V alternate power sources requirements of MIL-T-28800C.

3.5.2 Maximum power. The equipment shall consume not more than 150 watts.

3.5.3 Power connections. The equipment shall comply with the III power connections requirements of MIL-T-28800C except as specified herein.

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3.5.3.1 Power cables.

a. For 115 Vac operation, the power cables shall be in accordance with MIL-C-28777/3 except that the molded connector at the equipment end shall mate with Switchcraft Part Number EAC 302 connector or equal.

b. For 230 Vac operation the power cable shall be in accordance with W-C-596/17 at the source end.

3.5.4 Input power selection device. An input power selection device shall be provided for selection of input power voltages of 115 Vac or 230 Vac. Provision shall be incorporated to prevent accidental switching. The power selection device shall be mounted on the rear panel. When the equipment is delivered, the power selection device shall be in the 115 Vac position.

3.5.5 Fuses and circuit breakers. Fuses or circuit breakers shall be in accordance with MIL-T-28800C. (115 Vac/230 Vac) either common or separate fuseholders may be provided. If only one fuseholder is used (common), the equipment shall be provided with the 115 Vac fuse installed and the 230 Vac fuse shall be stowed with the accessories.

3.5.6 Input power switch. An input power switch need not be provided.

3.5.6.1 Standby-operate switch. A standby-operate switch or a power off-on switch shall be provided. The "STANDBY" or "OFF" position shall allow only the oven and its associated circuitry to be energized. The "OPERATE" or "ON" position shall allow all circuits to be energized.

3.5.7 Power indicator. A front panel light emitting indicator(s) shall be provided to indicate when the equipment is energized.

3.6 Dimensions, weight, and mechanical stability. The dimensions, weight, and mechanical stability shall be in accordance with MIL-T-28800C (see 4.5).

3.6.1 Weight. The weight of the equipment shall not exceed 15 Kg (33 lbs).

3.6.2 Dimensions. The dimensions of the equipment shall not exceed 483 mm (19.7 inches) wide by 140 mm (5.5 inches) high by 500 mm (19.7 inches) in depth. Frontal area shall not exceed 500 centimeters squared (80 square inches).

3.7 Enclosures. Enclosures shall be as specified in MIL-T-28800C for style E.

3.8 Marking and identification. The equipment shall comply with the type III marking and identification requirements of MIL-T-28800C. (See 4.5, 6.2.2.1 and 6.2.2.2). Nomenclature shall be assigned in accordance with MIL-N-7513.

3.8.1 System marking. The system part number shall be marked on the major assembly identification plate and on any externally removable subassemblies furnished in accordance with 3.4.2.

3.9 Environmental requirements. The equipment shall comply with class 5, style E environmental requirements of MIL-T-28800C, and as specified herein. (See 4.5 and 6.4.3).

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3.9.1 Electromagnetic interference (EMI). (See 4.5 and 6.4.3).

3.9.1.1 EMI. Unless otherwise specified in the contract or order, the equipment shall comply with the EMI requirements of MIL-STD-461B, part 2, class Alc, and as specified in a through g below.

a. CE03.

b. CE07.

c. CS02.

d. CS06.

e. RE02; relaxed 20dB, limit to 1 GHz.

f. RS02; parts I and Il.

g. RS03; 1 volt/meter limit to 1 GHz.

3.9.2 Warm-up. Warm-up shall be as specified in 3.12.2.

3.10 Reliability requirements. Each equipment delivered against this specification shall be subjected to a burn-in of not less than 100 hours at 40 degrees C of which at least the last 30 hours shall be failure free (4.4.6).

3.10.1 Each equipment shall be designed to have a mean time between failures of 2,000 hours.

3.11 Maintainability requirements. The equipment shall comply with the type III maintainability requirements of MIL-T-28800C and as specified herein (see 6.2.3.2).

3.11.1 Certification time. Certification time shall not exceed 2.0 hours, and it shall include all time required to check all inputs and outputs, on all range function settings.

3.11.2 Maintenance provisions. The equipment shall meet the maintenance provisions of MIL-T-28800C except the accessibility of accessories does not apply.

3.11.2.1 Maintenance and calibration aids. When required in the contract or order, any circuit board extenders, special adapters, special tools, and patch cables required for maintenance or calibration of equipment shall be stowed in a suitable reusable pouch or container which shall be marked "MAINTENANCE AIDS FOR USE WITH (equipment nomenclature or model number)" (see 6.2.2.1). The equipment nomenclature shall be used thereon if assigned (see 6.2.2.1).

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3.12 Performance requirements. The equipment shall contain the following electrical performance capabilities (see 4.5).

a.	Environmental performance	(3.12.1)
b.	Warm-up performance	(3.12.2)
c.	Time base	(3.12.3)
d.	Display	(3.12.4)
e.	Sample rate	(3.12.5)
f.	Self check	(3.12.6)
g.	Frequency measurements	(3.12.7)
h.	Input characteristics	(3.12.8)
i.	Outputs	(3.12.9)
j.	Logic interface	(3.12.10)
k.	Accessories	(3.12.11)

3.12.1 Environmental performance. Equipments shall meet the requirements specified herein under all combinations of environments specified herein, and with MIL-STD-210. The equipment shall also comply with MIL-STD-1568 for materials and processes to be as corrosion proof as possible in an aerospace environment.

3.12.2 Warm-up performance. Warm-up time shall not exceed one minute after the operate mode is selected when sufficient time has previously elapsed in the standby mode to permit the time base to be stabilized to the degree required for the measurement in accordance with 3.12.3.

3.12.3 Time base. Time base specifications are given in terms of fractional frequency error (ffe) (see 6.3.1). Except for temperature stability, all time base requirements are specified at 25 degrees C. Time base requirements are specified for measurement from the standard output with the equipment in the self test mode (4.5).

3.12.3.1 Long term stability (aging rate). The aging rate shall be less than $\pm 5 \times 10^{-10}$ per 24 hours after 72 hours of operation (Mg) preceded by at least 72 hours with no source power applied.

3.12.3.2 Frequency retrace. The ffe 30 minutes after application of source power (M2), following at least a 72 hour period of nonoperation, shall be within $\pm 2 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The ffe 5 hours after application of source/power (M_4), following at least a 2 hour period of nonoperation, shall be within $\pm 1 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The 72 hour period of nonoperation shall be preceded by at least 72 hours of operation. Refer to figure 1.

3.12.3.3 Restabilization. The ffe 5 hours after application of source power (M_6) shall be within $\pm 5 \times 10^{-9}$ of the ffe measured 48 hours application of source power (M_7). Application of source power shall be preceded by at least 72 hours of nonoperation.

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3.12.3.4 Short-term stability. The short term stability shall be less than $\pm 5 \times 10^{-10}$ root mean square (rms) 30 minutes after application of source power (M_3) and less than $\pm 5 \times 10^{-11}$ rms 5 hours after application of source power (M_5). Application of source power shall be preceded by at least 72 hours with no source power applied.

3.12.3.5 Stability during power change. The ffe shall not change more than $\pm 1 \times 10^{-8}$ during the first two minutes after any power change within the specified steady state band nor more than 1×10^{-10} at the end of the two minutes (see 3.5.1).

3.12.3.6 Stability during temperature change. The ffe shall not change more than $\pm 7 \times 10^{-9}$ total for any incremental or total temperature change within the specified temperature range (see 3.9).

3.12.3.7 Total ffe. The total calculated ffe at the end of a 12 month period shall not exceed $\pm 2 \times 10^{-7}$.

3.12.3.8 Oscillator adjustments. The oscillator shall be capable of being adjusted so that the 10 MHz standard output shall be within ± 1 part in 10^6 from 10 MHz.

3.12.3.9 External standard input. The equipment shall be capable of specified operation from any 10 MHz external frequency standard having a value of 1 volt rms into 50 ohms. When the external standard is used, it shall be in lieu of the internal standard.

3.12.3.9.1 External standard input impedance. The load impedance at the equipment external standard input port shall be not less than 1 kilohm paralleled by not more than 30 picofarad (pF).

3.12.4 Display. All front panel visible light or glow devices are display devices and shall be readable at a distance of 3 meters (9.9 feet) through a subtended angle of 45 degrees in an ambient light level of 100 footcandles. The numeric readout shall be an 11 character, inline, horizontal, single plane, digital display, at least 0.3 inches high. Provisions shall be incorporated to test all light emitting segments of the display devices.

3.12.4.1 Measurement units. Measurement units, in accordance with a through d, shall be presented in conjunction with the numeric display.

a. GHz

b. MHz

c. KHz

d. Hz

The LED display may be sectionalized into GHz, KHz, and Hz and appropriately labeled to eliminate shifting decimal point and digit overflow.

3.12.4.2 Display storage. The numeric readout shall be continuous and change only when the measurement is different from that counted during the previous measurement cycle.

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3.12.4.3 Display blanking. All non-significant leading zeros shall be blanked. All readout digits below the selected resolution of the displayed frequency shall not be displayed.

3.12.4.4 Gate lamp. A front panel display device shall indicate gate opening and closing.

3.12.4.5 Overload indicator. An overload display device shall warn of signal levels approaching the maximum operating input, at least 10 dB below damage level.

3.12.5 Sample rate. Except when logic interface is required, the sample rate shall be variable from 0.1 to at least 5 seconds. The sample rate shall be capable of being held indefinitely and shall be independent of the resolution control. When logic interface is required, the sample rates shall be as specified in 3.12.10.1.2. Erroneous data resulting from conflicts between sample rate and gate time shall not be displayed (see 6.3.3).

3.12.5.1 Reset. The control shall allow manual initiation of the rate.

3.12.6 Self-check. The self check function shall be provided to verify functional operation of the frequency measurement mode.

3.12.7 Frequency measurement.

3.12.7.1 Frequency range. The frequency range shall be at least 20 Hz to 26.5 GHz without external mixers. The equipment shall provide a feature by which a built-in frequency extender unit allows the counter to be used to measure signal frequencies to 40 GHz. This frequency extender shall provide the optional external mixers with an appropriate local oscillator (LO) signal from the internal VCO through the front panel connector for the first down conversion. The LO signal shall be separated from the IF signal received from the external mixer before supplying the down converted signal to the low impedance input of the counter. The frequency extender shall be capable of locking onto an incoming RF signal and tracking an RF signal once it is locked.

3.12.7.2 Frequency accuracy. The frequency measurement accuracy shall be $\pm$ one count $\pm$ time base stability.

3.12.7.3 Frequency resolution. Resolution from 1 Hz to 1 MHz in decade multiples shall be provided.

3.12.7.4 F.M. tolerance. The equipment shall tolerate and meet the accuracy requirements of 3.12.7.2 for signals between 500 MHz and 26.5 GHz which have peak-to peak deviation of up to 10 MHz at a deviation rate from zero to at least 10 MHz.

3.12.7.5 A. M. tolerance. The equipment shall tolerate and correctly measure an input signal which is a sine-wave, amplitude modulated with frequencies up to 1 KHz and has a modulation index of greater than 50 percent provided that the minimum signal level is not less than the sensitivity specifications.

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3.12.7.6 Signal discrimination. Discrimination between desired and undesired signals shall be provided in accordance with a or b.

a. The equipment shall lock onto and measure the signal which is 20 dB greater in amplitude than any other signal.

b. The equipment shall lock onto and measure the lowest frequency signal above the operator selected start frequency (comb line).

3.12.7.7 Measurement time. At one Hz resolution the equipment shall lock onto and measure the input signal in no more than 1.5 seconds.

3.12.8 Input characteristics. A maximum of three input ports shall be provided to cover the frequency range of 20 Hz to 26.5 GHz with a switching device for operator selection of the desired input port. An additional input port may be provided to incorporate the frequency extension feature (see 3.12.7.1).

3.12.8.1 Input impedance. The impedance of the high impedance port(s) shall be at least 1 megohm shunted by not more than 50 pF. The low impedance port(s), with the exception of the extended frequency port(s) shall have 50 ohms nominal.

3.12.8.2 Input port frequency range. The high impedance port(s) shall have a frequency range of at least 20 Hz to 20MHz. The low impedance port(s) shall have a frequency range of least from the upper limit of the high impedance port, with at least a 5% overlap, to 26.5 GHz without external mixers and to 40 GHz with external mixers.

3.12.8.3 Voltage standing wave ratio (VSWR). The VSWR of the 50 ohm input port(s) shall not exceed 2.5:1 up to 12.4 GHz; 3:1 from 12.4 to 18 GHz; and 5:1 from 18 GHz to 26.5 GHz.

3.12.8.4 Coupling. AC coupling shall be provided for the BNC input port(s), for the low impedance port(s) the coupling shall be dc to input load and ac to the instrument circuitry.

3.12.8.5 Sensitivity. The equipment sensitivity of the high impedance port(s) shall be at least 50 millivolts (mv), root mean square (rms). The equipment sensitivity of the low impedance port(s) shall be at least -25 dBm from 1 GHz to 18 GHz to 26.5 GHz, and -25 dBm from 26.5 GHz to 40 GHz utilizing external mixers.

3.12.8.6 Maximum operating input. The equipment shall operate satisfactorily with input signals of up to 120 volts rms at 1 kHz decreasing at a rate of 6dB per octave down to 2 volts rms on the high impedance input port(s) and up to +5 dBm on the low impedance input port(s) and the frequency extension port.

3.12.8.7 Maximum input without damage. The equipment shall accept without damage input signals of up to 150 volts rms at 1 kHz decreasing at a rate of 6 dB/octave down to 2 volts rms on the high impedance input port(s) and up to +25 dBm on the low impedance port(s).

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3.12.9 Standard frequency output. The equipment shall provide a 1 MHz or a 10 MHz standard frequency output which is independent of the front panel switch settings (except the standby-operator or power ON-OFF switch) and uninterrupted during all functions. The output signal characteristics shall be compatible with the external input requirements specified in 3.12.3.9.

3.12.10 Logic interface. A logic interface shall be provided in accordance with IEEE 488-1978.

3.12.10.1 Function programming characteristics. All measurement data, including multipliers, and units, shall be transferred by means of the logic interface. All control and switch functions except for the following shall be programmable.

a. Source power selection.

b. Frequency standard selection.

c. Input impedance selection.

d. Standby-operate (power ON-OFF) switch.

e. Sample rate potentiometer (analog function).

3.12.10.1.2 Sample programming. Under interface control the equipment shall have two sample modes (as selected by the controller). In the first mode the equipment shall not make any measurements until commanded to do so by the controller. The rate at which the controller can command the equipment to make measurements shall be at least 100 measurements per second or the maximum rate possible with required resolution. In the second mode, the equipment shall continually make measurements on its own at a rate of at least 100 measurements per second or at a maximum rate possible with the required resolution.

3.12.10.2 Interface characteristics. The equipment shall include the interface functions as in a thru f.

a. Source handshake interface function (SH), subset identification - SH1.

b. Acceptor handshake interface function (AH), subject identification - AH1.

c. Talker interface function (Y), subset identification - either T1 or T5.

d. Listener interface function (L), subset identification - L1, L2, L3, or L4.

e. Service request interface function (SR), subset identification - SR1.

f. Remote local interface junction (RL), subset identification - RL1.

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3.12.10.2.1 Numeric data. Unless otherwise specified in a contract or order, all numeric data (input and output) shall be American Standard Code for Information Interchange (ASCII) and the most significant digit shall be transferred first.

3.12.10.2.2 Alpha-numeric data. Unless otherwise specified in the contract or order, all alpha-numeric data (input and output) shall be ASCII.

3.12.10.2.3 Address assignment. Complete five bit switchable address capabilities shall be provided for the talker function (T1 or T5) address (MTA) and the listener function (L1, L2, L3, or L4) address (MLA) on the rear panel. If a single device for address switching is provided for both the MLA and MTA, then the five bit switchable address for the MLA shall be equal to that for the MTA.

3.12.11 Accessories. The following accessories shall be supplied with each equipment.

a. Power cable in accordance with 3.5.3.1.

b. One operator's manual. (Government approved)

c. Fuse in accordance with 3.5.5.

d. Frequency extension cable kit in accordance with 3.12.7.1.

3.12.11.1 Stowage. Accessories and maintenance and calibration aids (if required, see 3.11.2.1) shall be stowed in an external detachable pouch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Initial production articles testing will be performed by the Government.

4.1.1 Quality assurance terms and definitions. Quality assurance terms used herein shall be as in MIL-T-28800C. A Quality Assurance Program shall be established in accordance with MIL-Q-9858.

4.2 Classification of inspections. The inspections specified herein are classified as in a and b.

a. Initial production article inspection (see 4.3 and 6.2.1.6).

b. Quality conformance inspection (see 4.2.2).

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4.2.1 Initial production article inspection groups. Initial production article examinations and tests are classified into groups in accordance with table I and 4.3, and a through e below.

a. Initial production article inspection, group A (see 4.3.1.1).

b. Initial production article tests, group B (see 4.3.1.2).

c. Initial production article tests, group C (see 4.3.1.3).

d. Initial production article tests, group D (see 4.3.1.4).

e. Initial production article tests, group E (see 4.3.1.5).

4.2.2 Quality conformance inspection groups. Quality conformance examinations and tests are classified into groups in accordance with table I, 4.4, and a through e below.

a. Production inspection, group A (see 4.4.1).

b. Production control tests, group B (see 4.4.2).

c. Environmental and electrical power tests, group C (see 4.4.3).

d. Environmental tests, group D (see 4.4.4).

e. Other tests, group E (see 4.4.5).

4.3 Initial production articles instructions. Initial production articles instructions shall be as specified herein 4.3 and as specified in the solicitation.

4.3.1 Initial production articles inspection. Procurement of equipment by the initial production article method shall require the submission of a initial production article of the procuring activity when required in the solicitation. Unless otherwise required in the solicitation, the initial production article shall consist of two equipment with accessories and instruction manuals, a front panel sample (see 6.3.4), and clear and concise rationale showing how the reliability (see 6.4.1) and maintainability (see 6.4.2) and EMI characteristics (see 6.4.3) of the equipment complies with the requirements of the solicitation. The initial production article shall be representative of equipment specified by the requirements of this specification.

4.3.1.1 Initial production article inspection, group A. The two initial production article equipment will be subjected to the inspection specified in table I and II.

4.3.1.2 Initial production article tests, group B. The two initial production article equipment that have satisfactorily passed the required tests in accordance with 4.3.1.1 will be subjected to the group B tests of tables I and III.

4.3.1.3 Initial production article tests, group C. One initial production article equipment that has satisfactorily passed the required tests in accordance with 4.3.1.2 will be subjected to the group C tests of table I.

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TABLE I. Quality conformance inspection.

Examination test	Requirement Paragraph	Inspection Method	
		Initial production paragraph	Quality conformance paragraph
Group A			
Preoperational inspection	3	4.5	4.5
Level A performance	3.12	4.5	4.5
Leakage current	3.2	4.5	4.5
Group B			
Level B performance	3.12	4.5	4.5
Group C			
Temperature and humidity			
Humidity	3.9	4.5	4.5
Low temperature	3.9	4.5	4.5
High temperature	3.9	4.5	4.5
Temperature stability, time base	3.12.3.6	4.5	4.5
Altitude	3.9	4.5	4.5
Vibration	3.9	4.5	4.5
Shock			
Shock pulse	3.9	4.5	4.5
Electrical power			
Input power comsumption	3.5.2	4.5	4.5
Voltage and frequency			
Vibration	3.5.1	4.5	4.5
Voltage transient	3.5.1	4.5	4.5
Frequency transient	3.5.1	4.5	4.5
Line voltage stability time base	3.12.3.5	4.5	4.5
Verification, Group C	3.12	4.5	4.5
Group D			
EMI	3.9.1	6.4.3	4.5
Magnetic environment	3.9	6.4.3	4.5
Group E			
Maintainability	3.11	6.4.2	4.5
Accoustic noise	3.9	4.5	4.5
Dimensions	3.6	4.5	4.5
Weight	3.6	4.5	4.5
Mechanical stability	3.6	4.5	4.5
Front panel marking	3.8	4.5	4.5
Group F		6.4.1	

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TABLE II. Level A performance parameters.

Parameter	Requirement
Display	3.12.4
Sample rate	3.12.5
Self check	3.12.6
Frequency measurement	3.12.7
Input characteristics	3.12.8
Output	3.12.9
Logic interface (when required)	3.12.10

TABLE III. Level B performance parameters.

Parameter	Requirement paragraph
Warm-up	3.12.2
Time base	3.12.3
Time base, high stability	3.12.3.1
Long term stability	3.12.3.1.1
Frequency retrace	3.12.3.1.2
Restabilization	3.12.3.1.3
Short term stability	3.12.3.1.4
Stability during power change	3.12.3.1.5
Stability during temperature change	3.12.3.1.6
Total ffe	3.12.3.1.7
Oscillator adjustment	3.12.3.1.8
Time base, low stability (when required)	3.12.3.2
Long term stability	3.12.3.2.1
Frequency retrace	3.12.3.2.2
Restabilization	3.12.3.2.3
Short term stability	3.12.3.2.4
Stability during power change	3.12.3.2.5
Stability during temperature change	3.12.3.2.6
Total ffe	3.12.3.2.7
Oscillator adjustment	3.12.3.2.8
External standard input	3.12.3.3
External standard input impedance	3.12.3.3.1
Display	3.12.4
Numeric readout	3.12.4.1
Measurement units	3.12.4.1.1
Display storage	3.12.4.1.2
Display blanking	3.12.4.1.3

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TABLE III. Level B performance parameters - Continued.

Parameter	Requirement paragraph
Gate lamp	3.12.4.2
Overload indicator	3.12.4.3
Sample rate	3.12.5
Reset	3.12.5.1
Self-check	3.12.6
Frequency measurement	3.12.7
Frequency range	3.12.7.1
Frequency accuracy	3.12.7.2
Frequency resolution	3.12.7.3
F.M. tolerance	3.12.7.4
A.M. tolerance	3.12.7.5
Signal discrimination	3.12.7.6
Measurement time	3.12.7.7
Input characteristics	3.12.8
Input impedance	3.12.8.1
VSWR	3.12.8.2
Coupling	3.12.8.3
Sensitivity	3.12.8.4
Maximum operating input	3.12.8.5
Maximum input without damage	3.12.8.6
Standard output	3.12.9
Logic interface	3.12.10
Function programming characteristics	3.12.10.1
Sample programming	3.12.10.1.1
Interface characteristics	3.12.10.2
Numeric data	3.12.10.2.1
Alpha-numeric data	3.12.10.2.2
Address assignment	3.12.10.2.3

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4.3.1.4 Initial production articles tests, group D. Not applicable.

4.3.1.5 Initial production articles tests, group E. Equipment that have satisfactorily passed the required tests in accordance with 4.3.1.2 will be subjected to the group E tests of Table I. The number equipment tested will be as in a through e below.

a. Maintainability, unless otherwise specified in the solicitation, no equipment will be tested (see 6.4.2).

b. Acoustic noise. One equipment will be tested, except that equipment that do not contain a motor (such as a fan or blower) will not be tested.

c. Dimensions and weight. One equipment will be tested.

d. Mechanical stability. One equipment will be tested.

e. Front panel marking. One sample of the front panel will be tested (see 6.3.4).

4.4 Quality conformance inspection. Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as specified in 4.5 and MIL-T-28800C (see 6.2.2.7 and 6.2.2.8).

4.4.1 Production inspection. Each equipment shall be subjected to the group A inspection of tables I and II to ensure qualitatively the proper functioning of the equipment, including all operating controls, and conformance with safety requirement.

4.4.2 Production control tests. The group B tests of tables I and III shall be performed on equipment that have passed group A inspection (see 4.4.1). All test documents must comply with MIL-STD-1519.

4.4.2.1 Production control sampling plan. From each lot a random sample of equipment shall be selected and subjected to the group B tests. Minimum sampling shall be in accordance with lot size 20 quantity per lot 1.

4.4.3 Environmental and electrical power tests. The group C tests of table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.3.1 Environmental and Electrical power sampling plan. From each lot a random sample of equipment shall be selected and subjected to the group C tests. The sampling shall be in accordance with lot size 20 quantity per lot 1.

4.4.4 Enviornmental tests. The group D tests of table I shall be performed on equipment which has passed the required test in accordance with 4.4.2.

4.4.4.1 Environmental tests sampling plan. From each lot a random sample of equipment shall be selected and subjected to the group D tests. The sampling shall be in accordance with lot size, quantity per lot, as stated previously.

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4.4.5 Other tests sampling plan. From the first production lot, a random sample shall be selected and subjected to the group E tests. The sampling shall be in accordance with a through d.

a. Maintainability, one equipment.

b. Acoustic noise, one equipment.

c. Dimensions and weight, one equipment.

d. Mechanical stability, one equipment.

4.4.6 Reliability tests. No reliability testing is required. However, each equipment shall be subjected to a burn-in as specified in 3.10.

4.5 Examination and test methods. Examination and test methods shall be in accordance with MIL-T-28800C and as specified herein (see 6.2.2.6, 6.2.2.7, 6.2.2.8, 6.2.3.2, 6.2.3.3, tables I, II, and III). The time base testing shall be in accordance with figure 1. Equipment operation shall be verified in accordance with a and b below.

a. Satisfactory operation check. The satisfactory operation check shall confirm the proper operation of the internal time base, the input circuits of at least one channel, and the display section of the equipment.

b. Satisfactory operation test. The satisfactory operation test shall confirm the specified operation of the equipment for all parameters of the level A performance test (see table II).

4.5.1 Magnetic environment test. The magnetic environment test shall be performed in accordance with MIL-T-28800C. If the magnitude of the 20 oersted field is reduced as permitted in the test, the reduction may be to a 5 oersted level. Degradation of the equipment performance in a 5 oersted field or permanent damage to the equipment resulting from the 20 oersted field, shall constitute a failure of this test.

4.5.2 Alternate power source testing. Testing for the alternate power sources shall be in accordance with the electrical power tests of MIL-T-28800C except that alternate power shall be used.

4.6 Optional quality assurance testing. The procuring activity may substitute a Department of Defense (DoD) approved supplier's quality assurance system for any or all tests in table I.

5. PACKAGING

5.1 Preservation, packaging. Preparation of delivery shall be in accordance with MIL-T-28800C and MIL-E-17555 unless otherwise specified by the procuring activity.

6. NOTES

6.1 Intended use. Equipment described by this specification are intended for use in testing equipment and systems in the world-wide controlled and semi-controlled environments in which military equipment are operated (on a continuous or intermittent basis for extended periods of time), stored or transported

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6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

6.2.1.1 Title, number, and date of this specification.

6.2.1.2 Power cables if operation from 230V is desired (see 3.5.3.1).

6.2.1.3 Maintenance and calibration aids option (see 3.11.2.1).

6.2.1.4 Time base option if other than 3.12.3.1 (see 3.12.3.2).

6.2.1.5 Logic interface option (see 1.2 and 3.12.10).

6.2.1.6 Initial production articles.

6.2.1.7 Levels of packing and packaging (see 5.1).

6.2.2 Contract data requirements.

6.2.2.1 Nomenclature assignment (see 3.8 and 3.11.2.1).

6.2.2.2 National stock number (see 3.8).

6.2.2.3 Initial production articles.

6.2.2.4 Test equipment approval requests (see 4.5).

6.2.2.5 Pretest performance records (see 4.5).

6.2.2.6 Maintainability test records (see 4.5).

6.2.2.7 Quality assurance test plans (see 4.4).

6.2.2.8 Quality assurance test reports (see 4.4).

6.2.3.1 Reliability (see 3.10).

6.2.3.2 Maintainability (see 3.11).

6.2.3.3 Calibration system (see 4.5).

6.3 Definitions.

6.3.1 Fractional frequency error. Fractional frequency error is defined as the change (produced by any cause in frequency divided by the average frequency (before the cause). For example in determining the line voltage stability, if the standard output frequency were 1×10^7 MHz and if the standard output frequency varied by 10 Hz for any line voltage change within the specified steady state band then the ffe for the line voltage would be 10 divided by 1×10^7 which is equal to 1×10^{-6}.

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6.3.2 Short-term stability. The short term stability is defined as the square root of the Allan variance of the oscillator under test dividedby the average frequency of the oscillator under test. The band width shall be 30 KHz.

f_o = average frequency of oscillator under test.

$\bar{f}_k$ = the frequency sample (a frequency sample is a sample of the k frequency averaged over 1 second).

M = quantity of adjacent frequency samples.

(M = 40 for this specification)

$$\text{Short term stability} = \frac{1}{f_o}\sqrt{\frac{1}{2(M-1)}\sum_{K=1}^{M-1}\left(\bar{f}_{k+1} - \bar{f}_k\right)^2}$$

The potential vendor will be asked to supply test data on the 10 MHz time base standard in the counter demonstrating how it complies with 3.12.3.1 to 3.12.3.7. Additionally, a certificate of conformance to specification is required. This certificate must state that the crystal's calibration is traceable to the National Bureau of Standards.

6.3.3 Sample rate. Sample rate is the time interval between the trailing edge of one gate pulse and the leading edge of the next gate pulse.

6.3.4 Front panel sample. A front panel sample without holes, 100 mm (4 inches) X 100 mm (4 inches), marked with the same marking process to be used on production equipment, shall be included with the initial production articles (see 4.3.1).

6.4 Initial production article rationale.

6.4.1 Reliability rationale. The reliability rationale submitted with the initial production article, should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar equipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accompanied with specifications and technical literature on the similar equipment.

6.4.2 Maintainability rationale. The maintainability rationale submitted with the initial production articles should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design features relevant to maintainability may also be submitted (see 3.4). The

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acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production maintainability testing.

6.4.3 EMI rationale. The EMI rationale submitted with the initial production article, should provide clear and concise rationale basis showing how the EMI characteristics of the equipment comply with the minimum requirements of the solicitation. This may include data from previous EMI tests and/or predictions based on the design of the equipment, when such predicitions are fully explained and, where possible, backed up by test data obtained from testing done on similarily-designed equipment. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production EMI testing.

Custodians:
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Army - CR,
Navy - EC

Review activities:
Army - MI
Navy - SH, AS

User activities:
Navy - OS
Army - AR, SC

Preparing activity:
Air Force - 82

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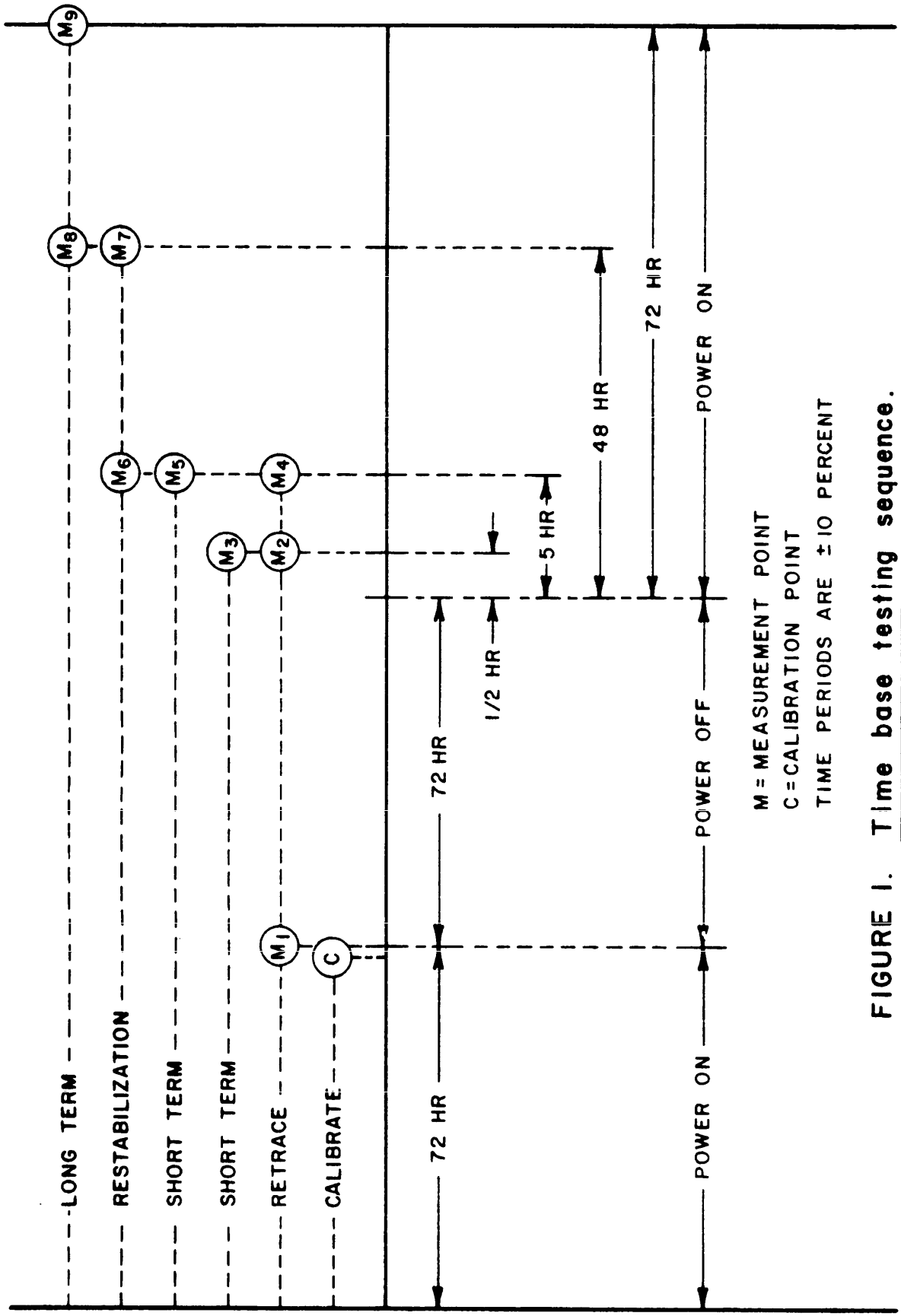


FIGURE 1. Time base testing sequence.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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